MATH6313 Lecture Notes

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1 Lecture 1 - 25/08/26

Calculus review.

2 Lecture 2 - 27/08/26

Theorem 2.1 (Taylor's theorem)**.** *Suppose $f \in C^n[a,b]$ that f^{n+1} exists on $[a,b]$ and $x_0 \in [a,b]$. $\forall x \in [a,b], \exists \xi(x)$ between x_0 and x s.t.*

$$f(x) = P_n(x) + R_n(x)$$

where

$$P_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2!}f''(x_0)(x - x_0)^2 + \cdots + \frac{1}{n!}f^{(n)}(x_0)(x - x_0)^n$$

$$R_n(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x - x_0)^{(n+1)}$$

3 Lecture 9 - 22/09/26

3.1 LU Factorization

Definition 3.1. Lower triangular matrix

$$\begin{pmatrix} x & 0 \\ x & x \end{pmatrix}$$

Definition 3.2. Upper triangular matrix

$$\begin{pmatrix} x & x \\ 0 & x \end{pmatrix}$$

Want to write the matrix as

$$A = LU$$

Example 3.3.

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2} & 1 \end{pmatrix}$$

Example 3.4. Define

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{pmatrix}$$

$$L_{21}(-3) = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$L_{32}(-2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$$

$$L_{21}(-3)A = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{pmatrix}$$

Repeat, which results in

$$L_{32}(-2)L_{21}(-3)A = \cdots = U$$

Now we get

$$A = LU$$

$$L^{-1} = L_{32}(-2)L_{21}(-3)$$

$$L = L_{21}^{-1}(-3)L_{32}^{-1}(-2) = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$$

3.2 Summary of LU Factorization method

Facts:

1. $L_{ij}(-c)$ denotes the lower triangular matrix whose only non-zero entries are 1's on the diagonal, and $-c$ is in the (i, j) position. Then $A \to L_{ij}(-c)A$ represents the RREF operation "row - row $\times c$".
2. $[L_{ij}(c)]^{-1} = L_{ij}(-c)$
3. From the previous example, $L_{21}(3)L_{32}(2) = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$ which gives us the full lower triangular matrix desired for LU factorization, which you can then use to find $U = L^{-1}A$.

3.3 Solving $Ax = b$

Given $LU = A$, you can solve $Ax = b$ by

1. $Lc = b$ (top to bottom)
2. $Ux = c$ (bottom to top)

Example 3.5.

$$LUx = Lc = b$$

$$\begin{pmatrix} 1 & 1 \\ 3 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -7 \end{pmatrix}$$

Using "$Lc = b$",

$$\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

solving from top equation to bottom equation yields

$$\Rightarrow \begin{cases} c_1 = 3 \\ c_2 = -7 \end{cases}$$

Now using "$Ux = c$"

$$\begin{pmatrix} 1 & 1 \\ 0 & -7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ -7 \end{pmatrix}$$

solve from bottom equation to top equation yields

$$\Rightarrow \begin{cases} c_1 = 2 \\ c_2 = 1 \end{cases}$$

3.4 Additional pivoting matrix P

If you have $PA = LU$, and you can solve $Ax = b$ by the following:

$$PAx = Pb$$

and thus $PA = LU$. Thus, you solve

$$\begin{cases} Lc = Pb \\ Ux = c \end{cases}$$

Example 3.6.

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 & 5 \\ 4 & 4 & -4 \\ 1 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 4 & 4 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & 8 \end{pmatrix}$$

Exercise to the reader.

3.5 Gaussian Elimination vs LU factorization efficiency

The computational resource for Gaussian is $\mathcal{O}(\frac{2}{3}n^3k)$, while the computational resource for LU factorization is $\mathcal{O}(2n^2k)$, where k is the dimension of the matrix.

3.5.1 Operation count for back substitution (LU method)

Suppose you have an $n \times n$ upper triangular matrix.

$$\begin{pmatrix} a_{11} & a_{12} \dots a_{1n} & b_1 \\ 0 & a_{22} \dots a_{2n} & b_2 \\ 0 & 0 \dots a_{nn} & b_n \end{pmatrix}$$

Then we solve

$$x_1 = \frac{b_1 - a_{12}x_2 - \cdots - a_{1n}x_n}{a_{11}}$$

$$\vdots$$

$$x_n = \frac{b_n}{a_{nn}}$$

So we have for computing x_1, 1 division operation, $n-1$ multiplications, and $n-1$ additions so we have $2n-1$ total operations. then for x_2, you have $2n-3$ operations, all the way until $1+3+5+\cdots+(2n-1) = n^2$ operations for x_n. So we have $\mathcal{O}(n^2)$.

3.5.2 Operation count for Gaussian elimination

Suppose you have an $n \times n$ matrix.

$$\begin{pmatrix} a_{11} & a_{12} \dots a_{1n} & b_1 \\ a_{21} & a_{22} \dots a_{2n} & b_2 \\ a_{n1} & a_{n2} \dots a_{nn} & b_n \end{pmatrix}$$

The function we want to write is:

```
for j = 1: n-1
    eliminate column j ⟶ zero below main diagonal
```

which is the same as

```
for j = 1: n-1
    for i = j+1:n
        eliminate a(i,j)
    end
end
```

explicitly, the algorithm is

```
for j = 1: n-1
    for i = j+1:n
        mult = a(i,j) / a(j,j)
        for k = j+1 : n
          a(i,k) = a(i,k)-mult * a(j,k)
        end
    end
end
```

To "eliminate a_{21}",

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ 0 & a_{22} - \frac{a_{21}}{a_{11}} a_{12} & \dots & a_{2n} - \frac{a_{21}}{a_{11}} a_{2n} & b_2 - \frac{a_{21}}{a_{11}} b_1 \\ a_{n1} & a_{n2} & \dots & a_{nn} & b_n \end{pmatrix}$$

which is 1 division, n multiplications, and n additions and subtractions. Thus we have a total of $2n+1$ operations. We have the following table for the number of operations for each matrix entry. The number of operations for each column decreases by 1 as you move left to right since we have one less entry to worry about.

$$\begin{bmatrix} 0 & & & \\ 2n+1 & 0 & & \\ 2n+1 & 2(n-1)+1 & 0 & \\ \vdots & \vdots & \vdots & \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & 0 \end{bmatrix}$$

Then the sum of total operations is

$$\sum_{j=1}^{n-1} \sum_{i=1}^{j} [2(j+1)+1] = \sum_{j=1}^{n-1} [2j(j+1)+j] = \mathcal{O}(\frac{2}{3}n^3)$$

3.6 $PA = LU$ factorization, or Matrix form of Gaussian elimination w/ partial pivoting

Example 3.7. Note, the fractions arent actually there, just to track for L.

$$A = \begin{pmatrix} 2 & 1 & 5 \\ 4 & 4 & -4 \\ 1 & 3 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 4 & 4 & -4 \\ 2 & 1 & 5 \\ 1 & 3 & 1 \end{pmatrix} \longrightarrow_{r_2 - \frac{1}{2} r_1, \quad r_3 - \frac{1}{4} r_1} \begin{pmatrix} 4 & 4 & -4 \\ \frac{1}{2} & -1 & 7 \\ \frac{1}{4} & 2 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 4 & 4 & -4 \\ \frac{1}{4} & 2 & 2 \\ \frac{1}{2} & -1 & 7 \end{pmatrix}$$

$$\longrightarrow_{r_3 - (-\frac{1}{2} r_2)} \begin{pmatrix} 4 & 4 & -4 \\ \frac{1}{4} & 2 & 2 \\ \frac{1}{2} & -\frac{1}{2} & 8 \end{pmatrix}$$

or equivalently

$$P_1 A, \quad P_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$M_1P_1A, \quad M_1 = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ -\frac{1}{4} & 0 & 1 \end{pmatrix}$$

$$P_2M_1P_1A, \quad P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$M_2P_2M_1P_1A = U, \quad M_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1}{2} & 1 \end{pmatrix}$$

thus combining we get

$$U = \begin{pmatrix} 4 & 4 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & 8 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 1 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

where P tracks the row swapping operations by multiplying the P_i matrices.
We can equivalently write

$$M_2P_2M_1P_1A = U$$

$$M_2P_2M_1P_2^{-1}P_2P_1A = U$$

$$M_2P_2M_1P_2^{-1}PA = U$$

$$L = (P_2M_1P_2^{-1})M_2^{-1}$$

Fact: $P_2M_1P_2^{-1}$ has same structure as M_1.

3.6.1 Summary

Definition 3.8. A permuatation matrix P is an $n \times n$ matrix consisting of all zeroes except for a single 1 in every row and column.

In general you have

$$M_{m-1}P_{m-1}\cdots M_2P_2M_1P_1A = U \iff M'_{m-1}M'_{m-2}\cdots M'_1P_{m-1}P_{m-2}\cdots P_1A = U$$

where $M'_{m-1} = M_{m-1}$, $M'_{m-2} = P_{m-1}M_{m-2}P_{m-1}^{-1}, \ldots, M'_1 = P_{m-1}P_{m-2}\cdots P_2M_1P_2^{-1}\cdots P_{m-1}^{-1}$.

4 Lecture 10 - 24/09/26

4.1 Iterative methods for $Ax = b$

Let x be the exact solution and the initial guess is x_0. Define $e := x - x_0$.

Definition 4.1. The residual is defined as $r := b - Ax_0$.

Notice that

$$Ae = A(x - x_0) = Ax - Ax_0 = b - Ax_0 = r$$

Thus, the error can be computed if the residual is known:

$$e = A^{-1}(b - Ax_0)$$

$$x = x_0 + A^{-1}(b - Ax_0)$$

however, we do not want to compute A^{-1}, so we will find something to replace it, giving the following iterative method:

$$x_{k+1} = x_k + B(b - Ax_k)$$

This turns out to be fixed point iteration:

$$x_{k+1} = G(x_k), \quad G(x) = x + B(b - Ax)$$

Write

$$A = L + D + U$$

where D is a strictly diagonal matrix and L, U as expected. We can then define $B := D^{-1}$ or $B := (L + D)^{-1}$.

4.1.1 Alternate derivation

$$Ax = b$$

$$(L + D + U)x = b$$

$$Lx + Dx + Ux = b$$

$$Lx_k + Dx_{k+1} + Ux_k = b$$

$$x_{k+1} = D^{-1}(b - Lx_k - Ux_k)$$

5 Lecture 11 - 29/09/26

5.1 Iterative Methods for Ax = b

Idea: say the exact solution is x and the initial guess is x_0. The error is $e = x - x_0$. The residual is $r = b - Ax_0$. We can relate the residual to the error by

$$Ae = A(x - x_0) = Ax - Ax_0 = b - Ax_0 = r$$

$$e = A^{-1}(b - Ax_0)$$

$$x = x_0 + A^{-1}(b - Ax_0)$$

The iterative method is then

$$x_{k+1} = x_k + B(b - Ax_k)$$

where B replaces the inverse of A^{-1}. Examples for B are

1. $B = D^{-1}$
2. $B = (D + L)^{-1}$

where $A = L + D + U$ where L is a lower matrix, D is a diagonal matrix, and U is a strictly upper matrix.

5.2 Jacobi Method

We take $B = D^{-1}$.

$$\begin{aligned} x_{k+1} &= x_k + D^{-1}(b - Ax_k) \\ &= x_k + D^{-1}(b - (L + D + U)x_k) \\ &= D^{-1}(b - (L + U)x_k) \end{aligned}$$

Which also can be written as

$$Dx_{k+1} = b - (L + U)x_k$$

or

$$Dx_{k+1} + (L + U)x_k = b$$

Example 5.1.

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$L = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Then we have the system

$$\begin{cases} 2x_{k+1} - y_k + 0 = 0 \\ -x_k + 2y_{k+1} - z_k = 1 \\ 0 - y_k + 2z_{k+1} = 0 \end{cases}$$

which solves to

$$\begin{cases} x_{k+1} = \frac{y_k}{2} \\ y_{k+1} = \frac{1}{2}(1 + x_k + z_k) \\ z_{k+1} = \frac{y_k}{2} \end{cases}$$

Definition 5.2. The $n \times n$ matrix $A = (a_{ij})$ is strictly diagonally dominant if for any $1 \leq i \geq m$, we have

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

Example 5.3.

$$\begin{pmatrix} 3 & 1 & -1 \\ 2 & -5 & 2 \\ 1 & 6 & 8 \end{pmatrix}$$

is strictly diagonally dominant, while

$$\begin{pmatrix} 3 & 2 & 6 \\ 1 & 8 & 1 \\ 3 & 2 & -2 \end{pmatrix}$$

is NOT strictly diagonally dominant.

Theorem 5.4. *If A is strictly diagonally dominant, then A is non-singular ($Ax = b$ has a solution). Additionally, for every b and initial guess x_0, the Jacobi method converges to the unique solution. This also applies to the Gauss-Seidel method.*

5.3 Gauss-Seidel method

For this method, we have $B = (L + D)^{-1}$.

$$\begin{aligned} x_{k+1} &= x_k + (L + D)^{-1}(b - Ax - k) \\ &= x_k + (L + D)^{-1}(b - (L + D + U)x_k) \\ &= (L + D)^{-1}(b - Ux_k) \end{aligned}$$

which can be written as

$$(L + D)x_{k+1} + Ux_k = b$$

Example 5.5.

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$L = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Then we have the system

$$\begin{cases} 2x_{k+1} - y_k + 0 = 0 \\ -x_{k+1} + 2y_{k+1} - z_k = 1 \\ 0 - y_{k+1} + 2z_{k+1} = 0 \end{cases}$$

$$L + U = \begin{pmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 0 & -1 & 2 \end{pmatrix}$$

which solves to

$$\begin{cases} x_{k+1} = \frac{y_k}{2} \\ y_{k+1} = \frac{(1 + x_{k+1} + z_k)}{2} \\ z_{k+1} = \frac{y_{k+1}}{2} \end{cases}$$

The $(L + D)^{-1}$ is exactly forward substitution method.

Lemma 5.6. *If A is strictly diagonally dominant, then* $\det(A) \neq 0$*, i.e. A is non-singular.*

Proof. If $\det(A) = 0$, then A has zero eigenvalues, i.e. $Ax = 0$ for some non-zero x. Then $\sum_{j=1}^{n} a_{ij}x_j = 0$ for all $i = 1, 2, \ldots, n$. Take the zero eigenvalue and set it equal to a_{ii}.

$$|a_{ii} \cdot x_i| = |a_{ii}||x_i| = \sum_{j \neq i} |a_{ij}x_j| \leq \max_{j \neq i} |x_j| \sum_{j \neq i} |a_{ij}| < \max_{j \neq i} |x_j| \cdot |a_{ii}|$$

$$\implies |x_i| < \max_{i \neq j} |x_j|, \quad \forall i = 1, 2, \cdots, n$$

which is a contradiction with s.d.d. □

Lemma 5.7. *If the $n \times n$ matrix A has spectral radius $\rho(A) < 1$ (all eigenvalues bounded above by 1), where*

$$\rho(A) := \max_{1 \leq i \leq n} \{|\lambda_i| : \lambda \text{ eigenvalue of } A\}$$

then for any b and x_0, $x_{k+1} = Ax_k + b$ converges. In particular, $\lim_{k \to \infty} x_k = x_*$ with $x_* = Ax_* + b$ and is unique.

Proof. Idea:

1. Uniqueness. Let $x_{*_1} = x_{*_2}$ such that $Ax_{*_i} = Ax_{*_i} + b$. Let $z_* = x_{*_1} - x_{*_2}$. Then $z_* = Az_* \implies 1$ is an eigenvalue of A, but $\rho(A) < 1$, so this is a contradiction.
2. Consider the geometric series $x_* = b + Ab + A^2b + \cdots$.

   $$x_* = Ax_* + b \iff (I - A)x_* = b$$

   this Neumann series converges due to $\rho(A) < 1$, thus a solution exists. Now, $e_k = x_k - x_*$.

   $$Ae_k = A(x_k - x_*) = Ax_k + b - (Ax_* + b) = x_{k+1} - x_* = e_{k+1}$$

   then

   $$\|e_{k+1}\| \leq \|Ae_k\| \leq \|A\|\|e_k\| \to 0$$

□

Proof of Theorem 3.4.

$$\begin{aligned} x_{k+1} &= x_k + D^{-1}(b - Ax_k) \\ &= (I - D^{-1}A)x_k + D^{-1}b \end{aligned}$$

By Lemma 3.7, it suffices to show that $\rho(I - D^{-1}A) < 1$.

$$(I - D^{-1}A)x = \lambda x$$

$$(D - A)x = \lambda Dx$$

$$(L + U)x = -\lambda Dx$$

$$|\lambda a_{ii}x_i| = |\sum_{i \neq j} a_{ij}x_j| \leq \max_{j \neq i} |x_j| \sum_{j \neq i} |a_{ij}| < \max_{j \neq i} |x_j| \cdot |a_{ii}|$$

Let $|x_k| = \max_{i=1,2,\ldots,n} |x_i| \neq 0$. Then

$$|\lambda a_{kk}x_k| \leq \max_{j \neq k} |x_j||a_{kk}| \leq |x_k||a_{kk}| \implies |\lambda| < 1 \implies \rho(I - D^{-1}A) < 1.$$

□

6 Lecture 12 - 01/10/26

Remark 6.1.

$$A = \begin{pmatrix} 1 & 4 \\ \frac{1}{8} & 1 \end{pmatrix}$$

is not strictly diagonally dominant, but $\rho(I - D^{-1}A) < 1 \implies$ convergence. So s.d.d. is a sufficient but not necessary condition.

6.1 Line search method

We will only deal with symmetric positive definite matrices.

$$A^T = A, \quad \forall x \neq 0,\ x^T A x > 0$$

Example 6.2.

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

is symmetric positive definite.

General method for line search method:

```
initial guess x_0
for k = 0, 1, ...
  choose search direction d_k
  choose step size alpha_k
  set x_(k+1) = x_k + \alpha_k * d_k
end
```

6.2 Steepest descent

We choose

$$d_k = r_k = b - Ax_k$$

$$\alpha_k = \frac{d_k^T r_k}{d_k^T A d_k}$$

Consider

$$f(x) = \frac{1}{2} x^T A x - b^T x$$

We want to minimize $f(x)$, so compute the gradient:

$$\nabla f(x) = Ax - b = -r$$

$$a_k = \min_{\alpha} \{f(x_k + \alpha d_k)\}$$

So to minimize,

$$\frac{\partial f}{\partial \alpha} = \nabla f \cdot d_k = [A(x_k + \alpha d_k) - b] \cdot d_k = (-r_k + \alpha A d_k) \cdot d_k = 0$$

$$(-r_k + \alpha A d_k) \cdot d_k = 0 \implies \alpha_k = \frac{d_k^T r_k}{d_k^T A d_k} = \frac{r_k^T r_k}{r_k^T A r_k}$$

Remark 6.3. We can use a better search direction, the conjugate gradient method.

6.3 Interpolation

Given a set of points, how can we find a polynomial that interpolates those points?

$$p(x_i) = y_i, \quad (x_i, y_i) \text{ are the given data points}$$

Theorem 6.4. *Given data points $(x_1, y_1), \ldots, (x_n, y_n)$, there exists a unique polynomial $p(x)$ of degree $n-1$ or less.*

$$p(x_i) = y_i, \quad i = 1, 2, \ldots, n$$

Goal: compute this polynomial $p(x)$.

Example 6.5. $(0,1), (2,2)$.

$$p(x) = a_0 + a_1 x$$

$$\begin{cases} a_0 + 0 * a_1 = 1 \\ a_0 + 2a_1 = 2 \end{cases} \implies \begin{cases} a_0 = 1 \\ a_1 = \frac{1}{2} \end{cases}$$

6.4 Lagrange Interpolation

Example 6.6. $(0,1), (2,2), (3,4)$. Define the following polynomials:

$$L_1(x) = \frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)}$$

$$L_2(x) = \frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}$$

$$L_3(x) = \frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)}$$

Then we can compute

$$L_1(x_1) = 1,\ L_1(x_2) = 0,\ L_1(x_3) = 0$$

$$L_2(x_1) = 0,\ L_2(x_2) = 1,\ L_3(x_3) = 0$$

$$L_3(x_1) = 1,\ L_3(x_2) = 0,\ L_3(x_3) = 1$$

Then we can compute the polynomial as

$$p(x) = y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x)$$

Definition 6.7. Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, define

$$L_1(x) = \frac{(x-x_2)(x-x_3)\cdots(x-x_n)}{(x_1-x_2)(x_1-x_3)\cdots(x_1-x_n)}$$

$$L_k(x) = \frac{(x-x_1)\cdots(x-x_{k-1})(x-x_{k+1})\cdots(x-x_n)}{(x_k-x_1)\cdots(x_k-x_{k-1})(x_k-x_{k+1})\cdots(x_k-x_n)} = \frac{\prod_{j\neq k}(x-x_j)}{\prod_{j\neq k}(x_k-x_j)}$$

These L_k are known as a Lagrange basis. then it follows

$$p(x) = y_1 L_1(x) + \cdots + y_n L_n(x) = \sum_{i=1}^{n} y_i L_i(x)$$

Question: suppose $y_i = f(x_i)$. We consider the interpolation error $f(x) - p(x)$.

7 Lecture 13 - 08/10/26

Remark 7.1. We have $L_k(x_k) = 1,\ \forall k = 1, 2, \ldots, n$ and $L_k(x_j) = 0,\ \forall j \neq k$

Theorem 7.2. *Assume $p(x)$ is the Lagrange interpolating polynomial that interpolates $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$. Define $f(x_i) = y_i$. Then the interpolation error is*

$$f(x) - p(x) = \frac{(x - x_1)(x - x_2)\cdots(x - x_n)}{n!} f^{(n)}(\xi)$$

where ξ lies between the smallest and largest of $x_1, x_2, \ldots, x_n$.

Proof. If $x = x_i,\ i = 1, 2, \ldots, n$, then we are done since it vanishes. If $x \neq x_i$, define for any t between the smallest/largest of x_i

$$G(t) = [f(t) - p(t)] - \frac{(t - x_1)(t - x_2)\ldots(t - x_n)}{(x - x_1)(x - x_2)\ldots(x - x_n)}[f(x) - p(x)]$$

notice that $G(x_i) = 0$ and $G(x) = 0$. By MVT, $G(t)$ has $n + 1$ distinct zeroes which implies $G'(t)$ has n distinct zeroes, which cascades to $G^{(n)}(t)$ has a single unique zero ξ. Thus,

$$G^{(n)}(t) = f^{(n)}(t) - p^{(n)}(t) - \frac{n!}{\prod_{i=1}^{n} x - x_i}[f(x) - p(x)].$$

Notice the $p^{(n)}(t)$ term vanishes because L has degree $n - 1$. Then we can compute

$$0 = G^{(n)}(\xi) = f^{(n)}(\xi) - \frac{n!}{\prod_{i=1}^{n} x - x_i}[f(x) - p(x)].$$

□

Example 7.3. $y_i = \sin(x_i)$, $x_1 = 0, x_2 = \frac{\pi}{6}, x_3 = \frac{\pi}{3}, x_4 = \frac{\pi}{2}$.

$$\sin(x) - p(x) = \frac{(x - x_1)(x - x_2)(x - x_3)(x - x_4)}{4!} f^{(4)}(\xi)$$

Example 7.4. Interpolate

$$f(x) = \frac{1}{1 + 12x^2}$$

at evenly spaced points in $[-1, 1]$. When computing, the error is large at the endpoints. This is known as the Runge phenomenon.

7.1 Chebyshev nodes and polynomials

We want a way to reduce this interpolation error and Runge phenomenon.

Theorem 7.5. *The choice of points $x_1, x_2, \ldots, x_n$ in $[-1, 1]$ that makes the values of the numerator of the interpolation error*

$$\max_{-1 \leq x \leq 1} |(x - x_1)\cdots(x - x_n)|$$

as small as possible is

$$x_i = \cos\left[\frac{(2i - 1)\pi}{2n}\right], \quad i = 1, 2, \ldots, n$$

They are called the Chebyshev nodes and the minimum value is $\frac{1}{2^{n-1}}$. The minimum is achieved by

$$(x-x_1)(x-x_2)\cdots(x-x_n)=\frac{1}{2^{n-1}}T_n(x)$$

where $T_n(x)$ denotes the degree n Chebyshev polynomial.

Remark 7.6. The Chebyshev nodes distribute in a way such that there are less points in the middle and more points near the endpoints of the interval.

Example 7.7. $f(x)=e^x$ with 5 points. $p_4(x)$ is the degree 4 Chebyshev interpolating polynomial.

$$f(x)-p_4(x)=\frac{(x-x_1)\dots(x-x_5)}{5!}f^{(5)}(\xi)$$

then

$$\begin{aligned}x_1&=\cos\frac{\pi}{10}\\x_2&=\cos\frac{3\pi}{10}\\x_3&=\cos\frac{5\pi}{10}\\x_4&=\cos\frac{7\pi}{10}\\x_5&=\cos\frac{9\pi}{10}\end{aligned}$$

We also know that

$$|(x-x_1)\cdots(x-x_n)|\le\frac{1}{2^{n-1}}=\frac{1}{2^4}$$
$$|f^{(5)}(\xi)|=|e^{\xi}|\le e$$

$$|f(x)-p_4(x)|\le\frac{1}{2^4}\cdot\frac{1}{5!}e\approx 0.00142$$

Definition 7.8. The nth degree Chebyshev polynomial is given by

$$T_n(x)=\cos(n\arccos(x)),\ -1\le x\le 1$$

Remark 7.9. We have the following recursive relation:

$$\begin{cases}T_0(x)=1\\T_1(x)=x\\T_{n+1}(x)=2xT_n(x)-T_{n-1}(x)\end{cases}$$

Proof.

$$T_{n+1}=\cos[(n+1)y]=\cos(ny+y)=\cos(ny)\cos(y)-\sin(ny)\sin(y)$$
$$T_{n-1}=\cos(ny)\cos y+\sin(ny)\sin(y)$$
$$T_{n+1}+T_{n-1}=2\cos(ny)\cos(y)=2xT_n(x)$$

□

Remark 7.10. The following facts are true:

1. $T_n(x)$ is a polynomial.
2. $\deg T_n = n$ with leading coefficient 2^{n-1} is monic.
3. $T_n(1) = 1$ and $T_n(-1) = (-1)^n$.
4. $\max_{-1\le x\le 1} |T_n(x)| \le 1$.
5. Roots of $T_n(x) = 0$ are

$$\begin{aligned} \cos(n \arccos(x)) &= 0 \\ n \arccos x &= \frac{\pi}{2}(2i-1) \\ x_i &= \cos \frac{\pi}{2n}(2i-1),\ i = 1, 2, \dots, n \end{aligned}$$

   where these x_i are the Chebyshev nodes.
6. $T_n(x)$ alternates between -1 and 1 a total of $n+1$ times.

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Proof of Theorem 7.5. We prove by way of contradiction. Suppose we have the polynomial

$$p(x) = (x - \hat{x}_1)(x - \hat{x}_2) \cdots (x - \hat{x}_n)$$

and

$$\max_{-1\le x\le 1} |p(x)| < \frac{1}{2^{n-1}}$$

We want to construct

$$Q(x) = p(x) - \frac{1}{2^{n-1}} T_n(x)$$

Since $p(x)$ is bounded above and below by $\pm\frac{1}{2^{n-1}}$ and $T_n(x)$ oscillates between -1 and 1 by Fact 3, $Q(x)$ alternates between negative and positive values $n+1$ times, thus $Q(x)$ has n distinct roots. But since $p(x)$ is monic with degree $n-1$ and $T_n(x)$ has degree $n-1$, $Q(x)$ has at most degree $n-1$. Thus, we have a contradiction. Therefore,

$$\max_{-1\le x\le 1} |(x - \hat{x}_1)(x - \hat{x}_2) \cdots (x - \hat{x}_n)| \ge \frac{1}{2^{n-1}}$$

for any $\hat{x}$.
If $x_i = \cos(\frac{2i-1}{2n}\pi)$, then

$$\max_{-1\le x\le 1} |(x - \hat{x}_1)(x - \hat{x}_2) \cdots (x - \hat{x}_n)| = \frac{1}{2^{n-1}} T_n(x)$$

and

$$\max_{-1\le x\le 1} |(x - \hat{x}_1)(x - \hat{x}_2) \cdots (x - \hat{x}_n)| = \max_{-1\le x\le 1} |\frac{1}{2^{n-1}} T_n(x)| = \frac{1}{2^{n-1}}$$

□

8.1 Cubic spline interpolation

Definition 8.1. Given n data points $(x_1, y_1), \dots (x_n, y_n)$ with strictly increasing x_i, a cublic spline $S(x)$ through the data points is a set of cubic polynomials:

$$\begin{aligned} S_1(x) &= y_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3 \text{ on } [x_1, x_2] \\ &\vdots \\ S_i(x) &= y_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 \text{ on } [x_i, x_{i+1}] \end{aligned}$$

for $i = 1, 2, \dots, n-1$.

Proposition 8.2. *The following properties hold for $S_i(x)$:*

1. $S_i(x_i) = y_i$ *and* $S_i(x_{i+1}) = y_{i+1}$
2. $S'_{i-1}(x_i) = S'_i(x_i)$
3. $S''_{i-1}(x_i) = S''_i(x_i)$
4. *The natural spline has the additional property of* $S''_1(x_1) = 0$ *and* $S''_{n-1} = 0$ *called end conditions.*

Proof. Property 1 direct computation:

$$y_{i+1} = y_i + b_i(x_{i+1} - x_i) + c_i(x_{i+1} - x_i)^2 + d_i(x_{i+1} - x_i)^3$$

The Property 2 and 3 computation is similar.
Property 4:

$$\begin{aligned} 2c_1 &= 0 \\ 2c_{n-1} + 6d_{n-1}(x_n - x_{n-1}) &= 0 = 2c_n \end{aligned}$$

Let $\delta_i = x_{i+1} - x_i$ and $\Delta_i = y_{i+1} - y_i$. this gives us a linear system

$$\begin{pmatrix} 1 & 0 & 0 & & \\ \delta_1 & 2\delta_1 + 2\delta_2 & \delta_2 & & \\ & \delta_2 & \ddots & \ddots & \ddots \\ & & \delta_{n-2} & 2\delta_{n-2} + 2\delta_{n-1} & \delta_{n-1} \\ & & & 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} 0 \\ 3(\frac{\Delta_2}{\delta_2} - \frac{\Delta_1}{\delta_1}) \\ \vdots \\ 3(\frac{\Delta_{n-1}}{\delta_{n-1}} - \frac{\Delta_{n-2}}{\delta_{n-2}}) \\ 0 \end{pmatrix}$$

□

Example 8.3. Fine the natural spline through $(0, 3), (1, -2), and (2, 1)$. We have $x_1 = 0, x_2 = 1, x_3 = 2, y_1 = 3, y_2 = -2, y_3 = 1$. Then $\delta_1 = 1, \delta_2 = 1, \Delta_1 = -5, \Delta_2 = 3$. Using the tridiagonal system above, we get

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 4 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 24 \\ 0 \end{pmatrix}$$

which yields

$$\begin{cases} c_1 = 0 \\ c_2 = 6 \\ c_3 = 0 \end{cases}$$

Then we solve for d_i and b_i using the values solved for c_i. Then you end with

$$\begin{cases} S_1(x) = 3 - 7x + 2x^3 \text{ on } [0,1] \\ S_2(x) = -2 - 1(x-1) + 6(x-1)^2 - 2(x-1)^3 \text{ on } [1,2] \end{cases}$$